Quantitative Methods

Module 8: Transportation Model

Name (LN,FN,MN): GARDNER, EUNICE B,   Program/Yr/Block: BSIT 2B _____

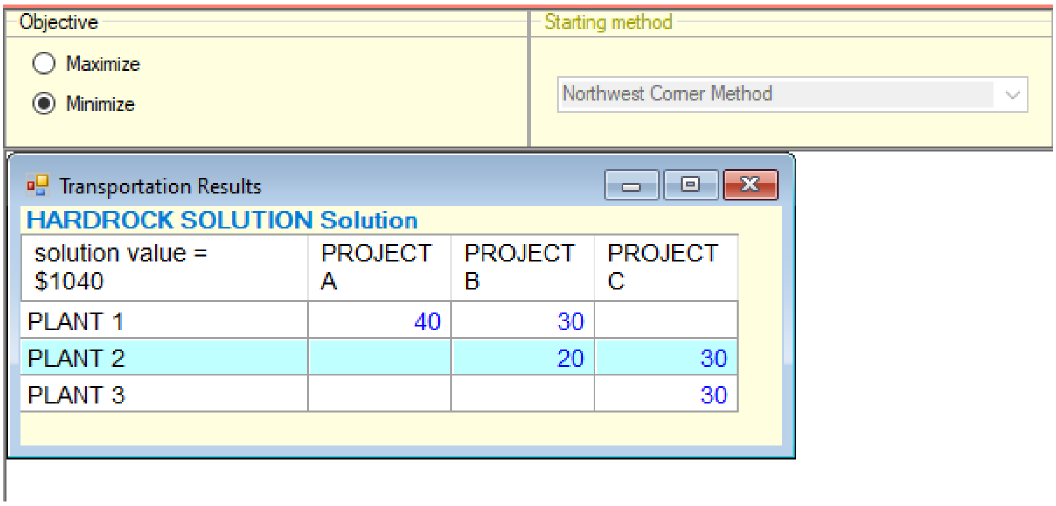
A. Explore

a. Use MS Excel QM/POM with the data given below to calculate transportation using northwest corner method. Take a screenshot of the output and paste it inside the box. (15 pts.)

Don Yale, president of Hardrock Concrete Company, has plants in three locations and is currently working on three major construction projects, located at different sites. The shipping cost per truckload of concrete, plant capacities, and project requirements are provided in the accompanying table. *(minimize cost)*

FROM \ TO	PROJECT *A*	PROJECT *B*	PROJECT *C*	PLANT CAPACITIES
PLANT 1	$10	$4	$11	70
PLANT 2	$12	$5	$8	50
PLANT 3	$9	$7	$6	30
PROJECT REQUIREMENTS	40	50	60	150

OUTPUT SCREENSHOT

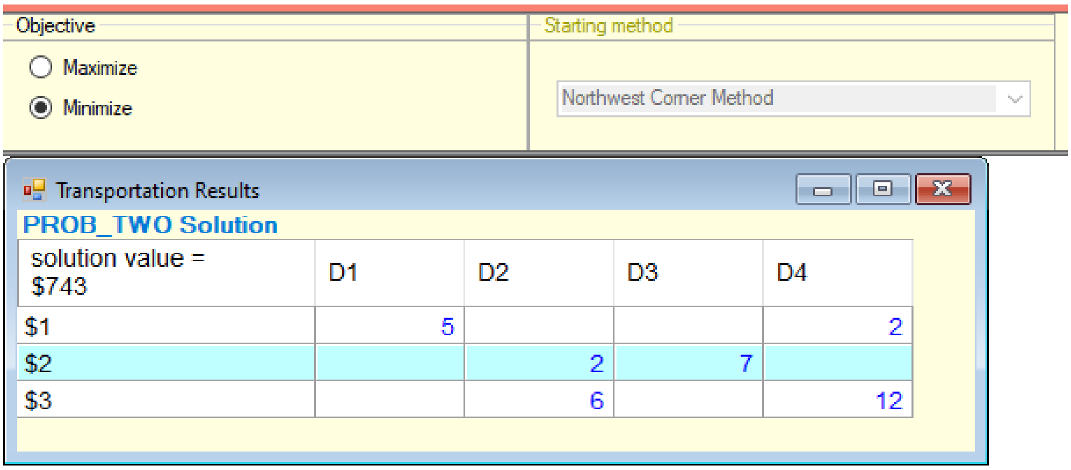


solution value = $1040	PROJECT A	PROJECT B	PROJECT C
PLANT 1	40	30	
PLANT 2		20	30
PLANT 3			30

b. Use MS Excel QM/POM with the data given below to minimize cost using northwest corner method. Take a screenshot of the output and paste it inside the box. (15 pts.)

	D1	D2	D3	D4	Supply
S1	19	30	50	10	7
S2	70	30	40	60	9
S3	40	8	70	20	18
Demand	5	8	7	14	34

Output Screenshot



B. Explain

a. Is the transportation model an example of decision making under certainty or decision making under uncertainty? Why? (15 pts)

I think the transportation model is an example of decision making under certainty because all the data is already given. Therefore, it is predictable, and the results can be seen accurately. There is no gamble in this model because the information is fixed and complete. Since nothing is missing and everything is provided, it clearly falls under decision making under certainty.